THE REPUBLIC OF UGANDA

MINISTRY OF WORKS HOUSING & COMMUNICATIONS
(MOWHC)

ROAD AGENCY FORMATION UNIT (RAFU)

PROCEDURAL GUIDE TO ECONOMIC ROAD FEASIBILITY STUDIES

Updated, March 2006

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Appendices

PREFACE

At the time of updating this Economic Feasibility Study Procedural Guide (March 2006), a new, revised version of the HDM-4 modelling software – version 2.0 – has recently become commercially available in late 2005.

The relevant web link (**HDMGlobal – managing HDM-4 Version 2**) is as follows: http://civ-hrg.bham.ac.uk/isohdm/

The improved version 2.0 contains a number of additional features in the model, however the model structure remains largely the same. These additional features include greater detail in the **configuration** definitions and options and the inclusion of parameters not previously incorporated within the modelling process. These new parameters relate to:

- Traffic Accident analysis
- Sensitivity analysis
- Multi-criteria analysis
- Estimation of social benefits
- Asset valuation

Much of what is written in this guide nonetheless remains relevant, however as the revised and updated HDM-4 v.2 model is used, this guide may need suitable revision and amendment in the future.

1.0 Introduction

1.1 The feasibility study is one of the stages comprising what is professionally known as the "project cycle" framework. This framework involves various steps from first identifying a project through (ex-ante) feasibility and engineering design stages to actual implementation, operation and (ex-post) monitoring/evaluation. The feasibility study stage, which is made up of a technical appraisal that usually involves a combination of economic, engineering and socio-environmental aspects, is required in order to assess whether the project should be carried forward for more detailed investigations and planning.

1.2 Given a situation of budgetary constraints, where there is competition for limited financial resources from different sectors within the national economy[1] as well as from different subsectors within the transport sector itself, the appraisal is, for the most part, based on economic decision criteria which rely on quantitative-based analytical methods.

1.3 In the case of proposed investments in road projects, the feasibility study may be applied to both paved and unpaved roads, encompassing:

For paved (sealed) roads:
- strengthening, rehabilitation or reconstruction;
- upgrading to higher standards in terms of geometric classification and/or pavement strength
- new road alignments

For unpaved (unsealed) roads:
- upgrading to paved road standard;
- improvement to higher gravel road standard;
- upgrading from earth tracks to gravel standard;
- new penetration road construction

1.4 The purpose of this procedural guide is to ensure a consistency of approach and applied methodology, in respect of the **economic analysis**, to the numerous road feasibility studies that are being carried out across various international donor agencies in Uganda[2].

[1] Health, education, public housing, public utilities such as water and electricity

[2] A very useful general guide to road project appraisal in developing countries is the document "Overseas Road Note 5 – A guide to road project appraisal" published by the Overseas Centre of the Transport Research Laboratory (TRL) in the UK. The first version of this document was published in 1988 with a new version of the guide (2005) recently becoming available. The relevant DFID (Department for International Development in the UK) web link is as follows:
http://www.transport-links.org/transport_links/news/news_v.asp?title=Overseas+Road+Note+5&id=230 or
http://www.transport-links.org/transport_links/filearea/publications/1_851_ORN_5_Final.pdf

2.0 Approach

2.1 The quantitative-based analysis is underpinned predominantly by four key variables:

- motorised vehicular traffic;
- vehicle operating costs;
- physical road characteristics;
- road maintenance and construction costs

Although traffic related benefits – in terms of vehicle operating cost savings and travel time savings – usually constitute the bulk of benefits accruing to a road project, besides those associated with road maintenance cost savings, there may be occasions when other project benefits are included in the analysis depending on the particular study in question and the availability of reliable data.

2.2 The subsequent sections of this guide examine separately, in greater detail, these different variables together with the analyses incorporated within the interactive modelling process, as follows:

- Section 3: Appraisal Methodology – Modelling
- Section 4: Traffic Studies
- Section 5: Other Project Benefits
- Section 6: Vehicle Operating Costs
- Section 7: Physical Road Characteristics
- Section 8: Road Maintenance and Construction Costs
- Section 9: Economic Appraisal
- Section 10: Sensitivity and Risk Analysis

3.0 Appraisal Methodology - Modelling

3.1 As part of the development of computer programs to model the interaction of traffic and road deterioration, various vehicle operating cost relationships have been derived from extensive research carried out over the past 30 years by both the World Bank and the Transport Research Laboratory (TRL) of the UK[3]. This research effort led to the development of detailed mechanistic relationships for predicting vehicle operating costs as a function of road pavement performance allied with vehicle characteristics. Whilst these relationships can readily be applied in road feasibility studies by means of spreadsheet analyses, it has now become common practice to utilise one of the well known and well established interactive road transport investment models, in particular the HDM (*Highway Design and Maintenance Standards*) model developed by the World Bank.

3.2 Historically, the HDM-III model, either the original 1989 version or the user-friendly HDM-Manager 3.0 1995 version (HDM-MAN), has been the application commonly used in international donor funded road feasibility studies, with the HDM-MAN version having been used in the (1999) updating of Uganda's first Ten Year Road Sector Development Programme (1996/97 – 2005/06). The revised and updated HDM-4 software (2000-2005), re-designated *Highway Development and Management*, has now superseded the earlier models.

3.3 In order to ensure a consistent approach to the various road feasibility studies that are carried out in Uganda and to enable such studies to be reviewed on a comparative basis across different donors, ***it is recommended that the HDM-4 model be used in the economic appraisal of road schemes***.[4]

3.4 The HDM-4 program models the relationships between traffic and pavement deterioration and maintenance over the design life of a road, within a project-based life cycle approach. The cost benefit analyses for the proposed road improvement alternatives are based on a comparison of the cost streams for the existing, *without project* situation – usually termed the *base case* – and the proposed *with project* road upgrading/improvement situation. The principal cost components incorporated in the analyses are the road construction costs and the road maintenance costs over the life of the project, and the costs of vehicle operation and travel time.

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3 Research carried out in Kenya (TRL), Brazil, India and the Caribbean (World Bank) in the 1970s and further research work carried out over the past decade that has included contributions from specialised road research organisations in Scandinavia, Australasia and other countries.

4 Where appropriate, for low volume road schemes (typically less than 200 vehicles per day), the related and less data intensive RED model – Roads Economic Decision Model, itself a spreadsheet type model based on the HDM relationships – can be used. Version 3.1 (August 2003) of the model can be downloaded from the World Bank website (www.worldbank.org) in particular the following web link: http://www4.worldbank.org/afr/ssatp/Resources/HTML/Models/RED_3.2/red32_en.htm

A new version 3.2 was released in July 2004. Refer to the following World Bank web link: http://www4.worldbank.org/afr/ssatp/Resources/HTML/Models/RED_3.2/english/red_documents.htm

3.5 The HDM-4 model structure comprises defining input data for the following key modules:

- *Vehicle Fleet*(s);
- *Road Network*(s);
- *Works Standards*, comprising
    - Maintenance standards;
    - Improvement standards; and
    - New construction sections (as relevant)

and definition of the particular type of analytical evaluation or appraisal to be conducted in terms of:

- *Projects* (study analysis of an individual road link(s) or section(s)); or
- *Programs* (study analysis of a road network comprising a number of separate road links and road classes); or
- *Strategies* (broad brush national or regional level analysis of road networks or road corridors)

One other module, in addition to the above set, is also a key module that should first be defined when setting an HDM-4 run. This module is:

- *Configuration* – that component of the model in which the basic, foundational aspects of the model are initially defined for the following parameters:
    - Traffic Flow patterns
    - Speed Flow types
    - Accident classes
    - Climate zones
    - Currencies
    - Road Deterioration Calibration sets (for different pavement types)

3.6 The first of the modules describes the characteristics and the price/cost values of the vehicle fleet together with the traffic growth sets for each of the vehicle classes. For those road sections of the road network comprising the analysis, the second of these modules defines their existing functional, physical, geometric and structural characteristics together with the associated average annual daily traffic (AADT). The third module describes the types of existing and proposed maintenance/improvement works allied with their unit costs. Lastly, one of the three following modules – most often, a project analysis of a specific road improvement or upgrading – describes the alternatives to be compared in terms of defining the base case alternative, which constitutes the existing road alignment, and each of the proposed road section alternatives, representing the new upgrading/improvement, with respect to their road engineering characteristics and traffic. The basic parameters concerning the analysis period and rate of discount to be used in the evaluation are also defined within one of these analysis modules.

3.7 The HDM model operates internally at an Information Quality Level, IQL-2 level[5]. In the hierarchy of road management data inputs to the model, this second level of information quality represents a relatively detailed amount of information in which the model has been structured to provide default values across a range of both road user effects (RUE) and road deterioration and works effects (RDWE) variables. Some degree of calibration is nevertheless required in order to tailor the model to local conditions, the level of which will be a function of the extent of resources – in both time and human capital – available for a particular study.

A Level 1 calibration – the basic application – is the minimum level required in any HDM application. Besides adopting many of the program's default values, this level determines the values of the model's basic input parameters with respect to the key variables within each of the Vehicle Fleet, Road Network and Works Maintenance and Improvement Standards modules highlighted in 3.5 above:

- Road pavement characteristics;
- Representative vehicle characteristics;
- Traffic composition and growth rates;
- Unit costs (for both the RUE and RDWE modelling components);
- Environmental (climatic) conditions.

5 Refer to Volume Five of the Highway Development and Management Series (Version 1.0, 2000): "A Guide to Calibration and Adaptation".

4.0 Traffic Studies

Traffic Surveys

4.1 Accurate traffic data is required for which it is necessary to carry out field surveys at selected sites on the study road in question. The surveys should involve:

- seven day, 12 or 16 hour traffic counts, usually 06h00-18h00, 07h00-19h00 or, security considerations permitting, 06h00-22h00;
- vehicle classification counts distinguishing between the following vehicle types having at least four wheels (approximate definitions given in brackets):

- car (sedan, saloon, estate);
- utility vehicle (pick-up, four wheel drive);
- minibus or *"matatu"* (having a capacity of not more than 14 passengers);
- medium bus or *"coaster"* ((having a capacity of not more than 25 passengers);
- large bus or *"coach"* (with a capacity of not more than 50 passengers);
- light truck (2 axle, gross vehicle weight of not more than 5 tonnes);
- medium truck (2 or 3 axle, gross vehicle weight of between 7 and 10 tonnes);
- heavy truck (3 or 4 axle, gross vehicle weight of up to 20 tonnes);
- articulated truck (semi-trailer) (4 and more axles, gross vehicle weight of more than 20 tonnes).

4.2 Optionally, counts may also be taken of two wheeled motorcycle and pedal cycle traffic and of pedestrian traffic, particularly in the case of very low-volume rural feeder roads or tracks, or if a study's Terms of Reference (ToR) and objectives warrant modelling of non-motorised traffic in terms of other considerations such as road safety or urbanisation and congestion-related issues.

4.3 Night counts should also be carried out in order to obtain 24 hour count totals and night factors. These should be performed preferably on one weeknight and one on the weekend but as a minimum, on one neutral weeknight (Tuesday, Wednesday or Thursday).

4.4 In the case of very low traffic volume roads, where average daily traffic (ADT) is of the order of less than 50 vehicles per day (vpd), and depending on the number of traffic count sites, two or three day counts may suffice. In this case, these should be carried out on neutral weekdays, that is, on a Tuesday, Wednesday or Thursday. For rural feeder roads, one of the day counts should coincide with the relevant market day in the project area.

4.5 As far as possible, counts should be carried out in the neutral months of the year, avoiding times of school vacation, agricultural harvests and national holidays. In similar vein, counts should not be undertaken during the rainy season or periods of particularly heavy rainfall, which would have the effect of suppressing normal traffic movements in the case of non all-weather rural roads and tracks.

4.6 Care must be taken to ensure that traffic count sites are located outside of urban or built-up areas. This is in order to avoid the data from being inappropriately augmented by localised traffic. Count stations, in the case of built-up areas, should therefore be located on the approaches to towns. *It should be borne in mind that the traffic counts are meant to represent the average traffic flow over the entire length of the road section under analysis.*

Historical Traffic Data

4.7 Historical traffic data should be obtained, as far as possible, for the study road in question. This data is useful in discerning any possible trends in traffic volumes and the changing composition of the traffic over time. Where such time series data is available over a reasonable period of time, at least five years and preferably ten, they might prove suitable in guiding the estimation of future traffic trends and projections.

4.8 Where historical traffic count records are available, they may also prove useful in gauging the extent of likely or potential traffic that could be possible given similar economic conditions in the project area as those that prevailed previously[6]. This would be the case, for example, where traffic levels are currently suppressed in the study area due to a combination of factors arising from domestic insurgency, insecurity, economic mismanagement and so on.

Traffic Forecasting

4.9 Traffic forecasts and projections can be determined from a number of variables. Most commonly, these variables include:

- historic traffic count data for the study road in question as well as historic records on traffic counts on adjacent road sections and the surrounding road network;
- historic data on gross domestic product (GDP) disaggregated where possible to the region/study area in question;
- population census data.

In the absence of regional/district level information on economic activity and output, national data on GDP is conventionally used. Data on national GDP and economic trends and short-term outlook can be sourced from official Government publications produced by the Uganda Bureau of Statistics (UBOS) (in particular, the annual "Statistical Abstract" document and the quarterly issues of "Key Economic Indicators") as well as from the Bank of Uganda and the major commercial banks. Sectoral GDP trends – specifically for agriculture, tourism or cross-border trade – may be especially relevant to the study road under analysis.

4.10 Other sources of data can be researched from World Bank publications as appropriate[7].

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[6] By implication, traffic levels as an indirect measure of the demand for goods and services, also reflect on the level of economic activity in the project area/region. Historical traffic records could be compared with prevailing GDP rates for the same period.

[7] Staff Appraisal Reports, Project Appraisal Documents etc.

4.11 For traffic growth purposes, consultants are at liberty to determine, as well as justify, their own rates of traffic growth for the first ten years of the scheme evaluation period. In the case of appraisal periods extending beyond ten years[8], the traffic growth rate must not exceed three per cent (3%) per annum in all cases or scenarios that are being studied.

4.12 In certain instances, there may be a requirement to determine the future distribution of traffic on the basis of a computerised traffic model. This requirement is most often the case in urban or possibly peri-urban situations, or where the study area involves an interface between the urban road network and the project road in question. In such cases, the conventional approach is to apply a standard traffic modelling suite, the sophistication of which is determined by the complexity of the problem being studied. Generally, in road feasibility studies, the modelling requirements are relatively simple in that all-or-nothing equilibrium assignment techniques may be applied in the modelling process.

4.13 There are various types of traffic modelling software that are available. The suites vary in sophistication and complexity from the relatively simple QRS suite to the highly complex EMME/2, TRIPS and other such suites which are predominantly used to model congested urban road networks. Such models are land-use based in which the conventional four stage modelling process is applied. This four stage process comprises:

- trip generation;
- trip distribution;
- modal split; and
- trip assignment

Trip generation is based on existing and projected activities taking place in different land use zones whereas the trip distribution and modal split phases are determined by generalised trip costs, in terms of distance and time for each travel mode, between the different land use or ‘traffic’ zones. In the assignment phase, the traffic flows are assigned over the road network on the basis of various equilibrium assignment techniques ranging from all-or-nothing to incremental and stochastic methods.

4.14 The rationale of the modelling process is premised on reproducing the existing situation to a satisfactory degree of accuracy[9], after which the model is used to predict the future situation on the basis of forecasts in land use activities. An example of the appropriate application of a traffic model to a road feasibility study would be in the case of a by-pass of a large urban area.

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[8] A 20 year post construction/upgrading evaluation period is conventionally applied for road feasibility studies.

[9] By means of calibration and verification techniques using traffic count data over the road network being modelled. The accuracy level is usually of the order of +/- 10 to 20 percent of the actual (observed) traffic flows.

4.15 Other variables which may aid the traffic forecasting procedure include:

- vehicle usage in terms of time series data on vehicle kilometres and on the size of the vehicle fleet (vehicle licensing and vehicle registration records);
- time series data on fuel consumption records;
- price/income elasticities of demand for passenger transport with respect to bus travel;
- price elasticity of demand for freight transport;
- time series data on vehicle ownership;
- trends in national Gross Domestic Product compared with those for the transport and road sub-sectors.

4.16 Elasticities may be derived from relating the annual changes in real GDP[10] with the changes in vehicle registrations and licensing[11]. Other elasticities may also be derived by comparing the percentage year-on-year changes in real GDP for the economy as a whole with the changes in domestic product for the transport and road sub-sectors.

4.17 Fuel sales generally reflect petrol consumption in the case of light vehicles and diesel consumption in the case of goods and heavy vehicles. Notwithstanding efficiency improvements in fuel economy rates, historic fuel sales data may be a useful guide to national traffic growth trends.

4.18 For bus transport, elasticities explore the relationship between changes in price (bus fares) and income (using per capita GDP as the proxy) with changes in the number of bus passenger trips and bus passenger km travelled. Studies in developing countries conventionally indicate a price elasticity of demand ranging between –0.6 to –2.0 with an average of –1.0, that is, a 10 per cent fall in bus fares would lead to a rise in passenger demand and the number of passenger trips of between 6 and 20 per cent[12]. In the case of freight transport, the elasticity would explore the relationship between changes in freight charges with changes in the number of road freight km. In this instance, the price elasticity of demand for goods is influenced by the importance of transport costs in the price of the commodity concerned and elasticities are generally lower compared with passenger transport.

4.19 Time series data on vehicle ownership, if available, will also be a good guide to growth in the level of motorisation. *(To a limited extent and with reference to data given in the World Development Indicators (published by the World Bank), it may be possible to analyse comparative data on the relationship between vehicle ownership and per capita income, using GDP as the proxy for income. Using logarithmic regression analysis for multiple country data, trend lines can be derived indicating the potential for future growth in private car travel as low-income economies become wealthier.)*

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10 GDP at constant prices for a specified base year.

11 Caution must be exercised in the interpretation of new vehicle registration growth since there is little or no data on scrappage rates and the extent of second-hand purchasing.

12 Overseas Road Note 5, page 16, paragraph 3.25.

4.20 The estimates of traffic growth must also take into account the influence of any specific developments, or committed developments, in the project area. In particular, this should include any developments that have a relatively significant traffic generating or traffic attracting impact such as mining, tourism or commercial/industrial/agro-industrial facility. Consideration should also be given to the actual and projected economic life of such facilities and the particular type of vehicular traffic that is estimated to be generated or attracted.

4.21 In all instances, the derivation of traffic growth rates must be justified with reference to the assumptions on which they have been determined. The growth rates should, data permitting, distinguish between passenger (light vehicle) and freight (goods vehicle) trip making.

Types of Traffic

4.22 In the process of estimating future traffic that is expected to use the new/upgraded road facility, due caution must be exercised in dealing with the three different categories of traffic, namely

- normal traffic;
- diverted traffic; and
- generated traffic.

4.23 **Normal traffic** represents existing traffic that is currently using the road irrespective of any proposed road project. This type of traffic is adequately catered for by the traffic count surveys and the historic traffic data.

4.24 Depending on the structure and physical configuration of the road network, **diverted or redistributed traffic** may or may not be an issue. Where the road network is structured such that diverted traffic should be incorporated into the analyses (on the basis of time and distance savings afforded by the proposed project), origin destination (OD) surveys need to be carried out at strategic points and junctions on the road network. These should be conducted over a 12 hour period and may be either three or seven days in duration. If the existing traffic count data indicate significant volumes of night traffic, the OD survey, with the assistance of the Police, should also cover the period after dark. In the case of diverted traffic, consideration must be given to

- both short and long distance trips;
- traffic that is local, regional and international in nature including transit traffic; and
- possible diversion from other, non-road modes of transport.

4.25 **Generated traffic** represents altogether new trips, that is trips induced to use the new or improved facility as a result of time and distance cost savings or because of new infrastructural developments. Generated traffic can be composed of any combination of the following:

- travelling to new destinations for the same purpose as existing journeys;
- increasing the frequency of certain trips;
- making entirely new trips;
- new patterns of land use giving rise to new trip origins and/or destinations.

This category of traffic is typically the most difficult to quantify with reasonable accuracy and conventional practice in developing countries often uses a factor of 20 to 25 per cent of normal traffic as the estimated level of generated traffic in the opening year of the scheme. This factor is applicable to light vehicles only – cars, utility (pick-up) and four-wheel drive vehicles, and minibuses. The factor for light and medium trucks (of less than 10 tonnes GVW) is likely to be of the order of 10 per cent of normal traffic.

4.26 Generated traffic is also a reflection of the level of suppressed travel demand often due to poor accessibility and characterised by high cost, infrequent public passenger transport services where monopolies may exist or where passengers and freight share space on utility and goods vehicles. The incidence of generated traffic is most likely to occur in the case of new road construction and with reducing intensity as a road is upgraded from earth to all-weather gravel and from gravel to paved standards. The generated traffic impact of improvements to existing paved roads, where all-weather passability is already assured, is likely to be negligible.

4.27 Elasticities of the demand for transport may be used for estimating generated traffic. The estimation of generated traffic is based on estimating the shape of the demand function, which is expressed as an elasticity of demand. This shows the responsiveness of quantity demanded to a change in cost. In the case of generated traffic, the demand for trip making is related to expected reductions in journey costs and travel time. These may be expressed separately or as a generalised cost function (of both time and distance). An illustrative example is given below using a simplified model of the form:

$$\text{Generated Traffic Factor} = \beta\,[(Cijp / Cijb) - 1]$$

Where
β is the elasticity;
Cijp is the with project travel cost
Cijb is the base case, without project travel cost

Assuming:

Parameter	**Low**	**Medium**	**High**
β (elasticity)	-0.6	-1.0	-2.0
Cijp (time in minutes)	60	60	60
Cijb (time in minutes)	120	120	120
Generated Traffic Factor	**0.3**	**0.5**	**1.0**

In this example, savings in journey time only have been used. The generated traffic factor is the factor that is applied to normal traffic in the year of opening of the scheme. The principle applies equally to non time savings, that is to distance based vehicle operating cost savings[13].

[13] Savings in fuel consumption alone may be used to simplify the calculation.

The generalised cost approach would involve a combination of both time and distance based savings. Generated traffic factors may be derived by applying the elasticities to the estimated savings in public transport passenger fares and to the estimated savings in freight charges[14].

4.28 Where the base level of existing traffic is particularly low, indicating the likelihood of suppressed travel demand associated with problems of accessibility, it may be necessary to approach the issue of generated traffic from first principles. This is usually the case for public passenger transport. Here, the analysis should identify the population in the project road's area of influence together with the major market/trading days in the study area. Again, for illustrative purposes, the following example is shown below:

Study District/Region		**Project Road's Zone of Influence**	
Population	153 000	Zone of Influence in km^2*	987
Land Area in km^2	1 738	Population	86 888
Population Density (per km^2)	88	Average household size	5,6
		Number of Households	15 516
Household Demand for Transport		**Public Transport Supply**	
Number of market days per month	2	Minibus occupancy	12
Number of household trips to market	1		
Number of Trips and Demand for Minibuses			
Monthly two way household trips	31 031	* usually based on a quarter day's walk distance (in daylight hours, between ≈. 06h00-18h00) to/from road (approximately 10km either side of road depending on terrain) to allow time for travelling and trading/purchasing activities at market place.	
Annual household trips ('000)	372		
Annual minibuses required ('000)	31		
Daily minibuses	85		

Note: For public passenger transport, in the absence of any information, it would not be unreasonable to assume that one trip per household may be made once every two months to the nearest administrative centre for social/business purposes. Using the figures in the example illustrated, and assuming medium bus services with an average occupancy of say 27 passengers per bus were to ply the route, the average demand would be of the order of 10 medium buses per day. For conventional buses, at an average occupancy of twice that of a medium bus, the demand would be some 5 buses per day.

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14 For example, if bus fares are expected to fall by 20 percent as a result of savings in per km vehicle operating costs leading to more competition amongst bus operators and the provision of cheaper services, and given a price/ income elasticity of demand of -1.2, then the generated traffic factor for bus passenger trips would be (-0.2 x -1.2) = 0.24 i.e. generated traffic would be expected to be 24 percent of the normal traffic figure for bus trips in the opening year of the scheme.

5.0 Other Project Benefits

5.1 Where appropriate, consideration should be given to the quantification of benefits accruing to a road improvement project besides those arising purely from traffic related savings in vehicle operating costs. In the case of rural road improvement projects, non-traffic benefits are usually those associated with changes in agricultural production which are expected to be stimulated by the road development project.

5.2 For low volume roads where ADT is of the order of less than 50 vehicles per day (vpd), and where the primary economic activity is agriculture-based, the main focus of analysis for estimating road project benefits is on the agricultural production system and the projected changes in agricultural output. For higher levels of ADT, typically greater than 50 vpd, the prime emphasis lies on the analysis of road user benefits in terms of vehicle operating cost savings. Even in the case of road projects falling into this latter category, where traffic intensity is higher, the calculation of project benefits, whilst substantially road user based, can nonetheless be augmented by incorporating changes in agricultural performance provided that accurate and reliable data are available.

5.3 The method of analysis in this case is based on the value-added **agricultural producer surplus** approach[15]. The approach is based on a comparative analysis, over the project appraisal period, between the with project and without project situations, of the following key variables:

	Variable	**Unit**
(i)	Crop Area	Hectares
(ii)	Crop Yields	Tonnes per hectare
(iii)	Agricultural Output	Tonnes
(iv)	Local Consumption*	Tonnes
(v)	Exportable Surplus	Tonnes
(vi)	Farm-gate Price per tonne	Currency
(vii)	Production Value (Surplus)	Currency
(viii)	Production Costs	Currency
(ix)	Transport Costs to market	Currency
(x)	Agricultural Net Benefits	Currency

* this is a function of the per capita consumption per annum of the population falling within the influence area of the road project.

The analysis must be applied only to that land area falling within the zone of influence of the proposed road project. This is usually based on a quarter day's walk distance to and from the roadhead (half day in total) to allow time for travelling and trading/purchasing activities at the market place.[16]

[15] Carnemark, Biderman and Bovet: The Economic Analysis of Rural Road Projects, World Bank Staff Working Paper No 241, August 1976.

[16] Depending on the topographic nature of the terrain (flat, rolling, hilly, mountainous) and the produce and goods being carried, this may vary between 5 and up to 15 kilometres either side of the road (head-loading walk speed of 4-5 kph, mule or bicycle mode). "Walk contour" maps should show this area of influence.

5.4 The underlying rationale of the agricultural producer surplus approach may be summarised as follows:

The with road project situation, on the basis of improved accessibility enabling (a) both the cheaper transportation of agricultural inputs (fertilisers, improved seeds) and extension services (by agricultural extension officers to improve cropping practices), and (b) higher farmgate prices supported by the cheaper transportation of agricultural produce to market[17], assumes an increase in the amount of farmland under cultivation and an increase in crop yields for existing crops and/or changes in cropping practices from low value subsistence to high value cash crops. The same logic may also be applied, as appropriate, to livestock taking into account the carrying capacity of the land and the rate of domestic off-take for local consumption purposes.

5.5 Where the agricultural producer surplus approach is incorporated into the economic appraisal, care must be taken to avoid including vehicle operating cost savings accruing to agricultural-based vehicle traffic i.e. the vehicle operating cost calculations must distinguish between agricultural and non-agricultural freight traffic and exclude the former in the analyses, so as to avoid any incorrect double counting of benefits.

5.6 Besides benefits that may accrue from agriculture, the other principal source of exogenously determined benefits that may be included in the economic appraisal relates to the valuation of **traffic accidents**. In this case, it is necessary to obtain accurate historical data on traffic accidents, and types of accident, for the road section under study together with indicative traffic accident rates for different road types. To a degree, the analysis remains relatively subjective in determining accident rates for different types of road conditions and surfaces. It is nevertheless possible to quantify the economic costs of traffic accidents in terms of the severity of accident, this usually being classed into four categories: fatal, serious injury, slight injury and vehicle damage only.

5.7 Where serious human injuries and/or fatalities are involved, the principal method of analysis is based on the Gross Output or 'human capital' approach in which the value of human life is taken as the discounted present value of the accident victim's future economic output. The data requirements would include data on average incomes, as a function of per capita Gross Domestic Product (GDP, nationally or regionally), average economic working life[18] and other real resource costs such as those relating to medical, police and insurance costs.

5.8 As part of a Road Safety and Audit Improvement Study (RSAIS) carried out in 2000, an economic working paper was prepared that gave accident-related values using the Gross Output method[19]. The key variables and the relevant 1998 values from this study are given below.

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17 The mode of transporting produce to market may either change from non-motorised means such as head-loading/ donkey or bullock scotch-cart/ bicycle to motorised transport, or may improve through more frequent availability of utility vehicles (pick-ups) or light trucks.

18 The principle is to assess the loss of the potential (to the national economy) income-earning ability of the accident victim over an assumed average working life.

19 Phoenix Engineering & Research Ltd, April 2000. (Chapter 11, Appendix F4 of the RSAIS)

Fatalities

(a) Loss of Income

Average age of fatality (years)	0-16 age group: 12	17+ age group: 32
Beginning of working life (years)	17	17
End of working life (years)	60	60
Duration of remaining work life (years)	43	28
Average income per year*	US$ 250	US$ 250
Present value of lost input per worker killed**	US$ 4 349	US$ 4 124

* GDP per capita as proxy for income (together with estimated rate of real growth taking into account population and GDP time series data trends and projections – 1998 estimate: 3% p.a.)
** at 7% rate of discount over the remaining work life period

(b) Medical costs

Average number of days in hospital: 4	4
Average cost of hospitalisation per day	US$ 120
Percentage of fatalities in hospital	17%

Injuries

(a) Serious injuries

Average number of lost working days (income)	60
Average number of hospitalisation days	8

(b) Slight injuries

Average number of lost working days (income)	4
Average number of hospitalisation days	0.5

Vehicle (property) damage only

In the RSAIS (2000) study, an average damage only (1998) value of US$ 2 300 per accident was estimated. This figure must be appropriately updated through consultation with the major insurance and loss-adjuster companies.

Consideration must also be given to the extent of accident under-reporting. Under-reporting of slight injury and vehicle damage only accidents in the RSAIS study were estimated to be of the order of 1:2 (50%) and 1:8 (87.5%) respectively.

5.9 Road development projects may also result in other indirect economic benefits in the form of enhanced land values for properties adjacent to the improved or upgraded facility. Generally speaking, estimating the increase in property land values resulting from infrastructural developments is usually carried out in urban corridors in developed countries where land values are at a premium. This type of analysis is not normally included for road feasibility studies in developing country situations, except in cases of urban by-passes.

6.0 Vehicle Operating Costs

6.1 Vehicle operating costs (VOC) are one of the key components in road project appraisal at the feasibility study stage. Application of vehicle operating costs to the projected traffic flows over anything between a 10 to 30 year project appraisal period, in both the with project road upgrading or improvement situation and the without project existing road situation, provides the analytical basis for determining a specific road scheme's net benefits.

6.2 The level of calibration undertaken in Uganda has generally been at the minimum, basic Level 1 standard; however, some degree of Level 2 calibration was carried out as part of the National Transport Master Plan (NTMP) study that was conducted in 2003 and 2004.

6.3 VOC-related values reported on in the NTMP study are given in *Appendix A* of this Procedural Guide together with indicative values for new vehicle and tyre prices that have been applied in various road feasibility studies that were carried out between 2003 and 2005.

Notwithstanding the values given in *Appendix A*, VOC-related unit costs and prices must nevertheless be updated by researching the latest values in Kampala and taking into account any regional differences where appropriate.

6.4 Amendments or modifications must also be made to reflect the particular road feasibility study in question and where new and/or updated data become available, with reference to the following parameters:

- (i) vehicle occupancy and trip purpose information from roadside surveys;
- (ii) equivalent standard axle (ESA) factors from axle load surveys;
- (iii) where new/updated vehicle utilisation data is available from surveys or other studies in respect of annual hours and annual kilometrage;
- (iv) where the representative vehicle type(s) may change over time as newer models become popular;
- (v) values of travel time.

6.5 The estimation of passenger time values relies on the average income method and should reflect income per employed person, using GDP as the proxy for income rather than GDP per capita or per household. This value should then be factored by the proportion of GDP that represents consumption expenditure. In the absence of specific studies in developing countries in the region, the value of non-working time is generally of the order of between 25 and 30 per cent of the value of work time.

6.6 The price structure for fuel also needs to be analysed in order to determine the economic value for petrol and diesel. Starting with the CIF price at the port of entry (Mombasa in Kenya), this should distinguish between the financial and economic cost elements for each of the (i) haulage (by road transport and/or pipeline) to Kampala, (ii) the distribution cost within Uganda, (iii) the distributor margin, and (iv) the dealer margin components to arrive at the pump price.

7.0 Physical Road Characteristics

7.1 Whilst field surveys and measurements on road roughness should ideally be carried out during the feasibility study[20], subjective guidelines on estimating values of road roughness are given in *Appendix B*, based on the experience of the World Bank and the Transport Research Laboratory. Road condition surveys should, resources permitting, be carried out by driving over the full length of the study road with a visual estimation based on sound engineering judgement and experience. Such surveys should, preferably using vehicle-mounted GPS[21] equipment, record, on an 'x' kilometre chainage basis, the extent of the following key variables, in terms of subjective criteria ranges which would subsequently be separately defined in specific detail in preparing the input data for the HDM model:

- gradients – level/medium/steep;
- bendiness (curvature) – straight/gentle/sharp; and
- surface condition roughness – poor/fair/good

7.2 A caution needs to be raised concerning homogeneity. For appraisal purposes, it is necessary to ensure that each road section under study has both homogeneous traffic and homogenous physical road characteristics. Where traffic flows vary significantly between sections, usually between towns or larger settlements, between important junctions or where there are significant traffic generating developments (such as mines, factories or agro-industrial plants), the sections should be analysed separately.

7.3 Similarly, separate analyses of road sections should be conducted where the physical characteristics of the road sections are significantly different, particularly in respect of geometry and pavement structure. In the case of geometry, this refers to the carriageway width (effective number of lanes and shoulders), whilst in the case of pavement structure, distinctions must be made with reference to surface texture and condition (sealed/unsealed road, road roughness) as well as to the pavement layers and thicknesses for purposes of analysing paved (sealed) road pavement strength parameters[22].

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[20] by using a suitably calibrated Bump Integrator (BI) device

[21] Global Positioning Satellite

[22] resources permitting, pavement strength can be measured by using a Falling Weight Deflectometer (FWD) device and the pavement structure (types of materials/aggregates and layer thicknesses) can be sampled using coring techniques. (Application of these measuring/survey methods would constitute a Level 2 model calibration).

8.0 Road Maintenance and Construction Costs

8.1 Unit costs for road maintenance operations – routine and periodic – should be obtained from the maintenance division within RAFU / UNRA. These should be verified to ensure that they reflect the current costs of maintenance activities and modified for any regional disparities as appropriate. The definitions of the various types of road maintenance operations are defined by the requirements of the project and with reference to the HDM model.

8.2 Unit costs for new road construction and for road upgrading, improvement or reconstruction works should be obtained with reference to a review of the most recent tenders for road construction activity in Uganda. The derivation of construction costs should be based on an analysis of the quantities allied with the unit rates. For the purposes of the economic appraisal, the costs should be expressed in financial costs per kilometre, but based on the estimated quantities and unit rates from the engineering surveys and preliminary design studies.

8.3 The cost calculations should be based on an estimate of the standard work items and other cost-related contingencies itemised in the preliminary or detailed design engineering Bill of Quantities (if available), as follows:

- Preliminary and General
- Road Earthworks
- Pavement Construction
- Surfacing
- Culverts and Drainage
- Structures and Bridges
- Ancillary Works
- Dayworks
- Physical Contingencies (usually ten per cent)
- Construction Supervision Costs (usually seven per cent)
- Environmental Mitigation Costs
- Land Acquisition (compensation) Costs

8.4 Within the HDM model structure, different types of road maintenance and improvement works standards need to be defined for both the with and without project situations. These are to be determined by the analyst and should properly reflect the requirements and objectives of the feasibility study.

9.0 Economic Appraisal

9.1 Economic values should differentiate from financial prices by removing all transfer price elements such as taxes, import duties and levies, and subsidies. In Uganda, the open market economy coupled with the free movement of the country's exchange rate and the absence of any foreign exchange controls, means that the Ugandan Shilling is not overvalued with respect to international currencies. Similarly, there is no minimum wage laws or regulations.

9.2 The standard conversion factor (SCF), using the Little & Mirlees[23] formula, should be applied. This takes the form:

[Imports + Exports] / [Imports + Exports + import duties/taxes + export subsidies]

In the case of construction costs, an alternative approach would be to analyse the composition of construction costs at market prices and deduct actual tax and import duty rates applied to the individual cost components.

9.3 A residual value may be applied in the last year of the appraisal period to the built infrastructure. The impact of discounting means that the contribution of this residual value will be relatively marginal in the scale of total project benefits. A residual value will only be applicable to infrastructure that still retains an opportunity cost of capital beyond the design life of the road and the project evaluation period. The most obvious case would be a bridge structure whose design life may be 120 years[24]. In general, a residual value of a road is of the order of ten to twenty per cent, generally reflecting the value of the existing formation and base pavement layers.

9.4 The discount rate, which is used to convert all future costs and benefits to present day values, is a reflection of the opportunity cost of capital. In the case of road improvement/upgrading projects that are funded by international donors, the rate of discount is usually determined by the particular lending agency concerned. For World Bank and European Union funded projects, the discount rate has for many years conventionally been 12 per cent; however, rates of 10 per cent or even less may be used depending on the lending agency and the socio-economic circumstances of the particular road project in question.

[23] I. D.M. Little and J. Mirlees – Economic Analysis of Projects in Developing Countries (Oxford, 1974)

[24] British Standard BS 5400 (1974), Section 1.

9.6 The results of the economic appraisal are commonly expressed in the principal economic efficiency indicators of Net Present Value (NPV) and Economic Internal Rate of Return (EIRR). Other indicators include First Year Rate of Return (FYRR), Benefit Cost Ratio (BCR) and the ratio of Net Present Value to Cost (initial financial capital investment) (NPV/C). The relative merits of these indicators in terms of different project appraisal applications are shown below:

Summary of Economic Efficiency Indicators for Road Project Appraisal

Appraisal Option	**NPV**	**EIRR**	**FYRR**	**BCR**	**NPV/C**
Single Road Investment	Y	Y	Y	Y	
Multiple Road Investments (mutually exclusive) under Budget Constraint	Y	Y	Y		Y
Road Network Analysis under budget constraint (prioritisation/ranking)	Y		Y		Y
Phased Road Investment (project timing)	Y	Y	Y	Y	

10.0 Sensitivity and Risk Analysis

10.1 Sensitivity and risk analyses are conducted to test the robustness of the appraisal results to changes in the base case set of assumptions. Sensitivity analysis focuses on the impact of varying the magnitude of one of the significant variables in isolation; the emphasis in risk analysis on the other hand is on estimating the probability of variations occurring in one or more of the significant variables. The usual application in road feasibility studies is to vary the magnitude of road construction costs and to vary base year traffic and/or traffic growth rates.

10.2 For the sensitivity analysis[25], the variation in road construction costs is usually of the order of between plus and minus 10 to 20 or 25 per cent. In the case of traffic, two variations are applicable. On the one hand, the base year traffic level may be varied by plus and minus 10 to 20 per cent; and on the other, the rates of traffic growth may be varied by plus and minus 10 to 25 per cent, to reflect optimistic and pessimistic scenarios respectively. The worst case scenario – increased construction costs and reduced traffic in combination – is also to be reported on.

10.3 Another indicator that may also be used in reporting the sensitivity of the appraisal results is switching-value analysis. This type of analysis reveals the extent to which the key input variables (construction costs and traffic level) would need to change (or switch) for the project to be just viable at the specified rate of discount (i.e. when IRR = 12% or when NPV = 0 at the 12% discount rate). Switching values are a measure of the robustness of the key input variables – in terms of, respectively, the relative extent of increase (percentage rise) in construction costs and the relative extent of reduction (percentage decline) in traffic benefits.

10.4 Risk analysis is concerned with the probability or likelihood of construction costs increasing by a certain percentage and/or traffic growth occurring at a lower rate than that assumed in the base case situation. The principle of risk analysis is based on random simulations using the Monte Carlo or Latin Hypercube sampling types in standard risk analysis software programs[26]. There are two such programs available, @RISK (which is Excel spreadsheet based) and Crystal Ball. The simulations are carried out assuming a triangular probability distribution with predetermined minimum, most likely and maximum limits. The most likely values are those assumed for the base case situation, whilst the lower and upper limits that are to be applied for risk analysis in road feasibility studies are as follows:

- 90% and 120% of the base case road construction costs;
- 75% and 110% of the base case traffic growth rates.

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[25] In the new version 2.0 of the HDM-4 model, this is now one of the areas that represents a significant improvement to the earlier versions. Previously, sensitivity could not be modelled in version 1.3 (although the earlier HDM-MAN 1995 DOS-based model did provide this facility). The latest version of the model permits a wide variation of sensitivity tests to be carried out within three main parameters: traffic, vehicle use and net benefits. Refer to the following web link: http://civ-hrg.bham.ac.uk/isohdm/ (*Summary of Improvements in HDM-4 Version 2.0.*)

[26] The simulation process simultaneously analyses every possible outcome from hundreds of "what if" scenarios, or user-specified number of iterations.

11.0 Checklist

Road Feasibility Study **CHECKLIST**
of Inputs to, and Analyses of, the Economic quantitative-based methodology:

<table>
<tr><th colspan="3">INPUTS</th><th colspan="2">ANALYSES</th></tr>
<tr><td>A</td><td>Road Network/Section(s) definition</td><td>Mandatory</td><td colspan="2" rowspan="2"></td></tr>
<tr><td>B</td><td>Vehicle Fleet definition</td><td>Mandatory</td></tr>
<tr><td>C</td><td>Traffic Volume</td><td>Mandatory</td><td>H</td><td>Interactive Modelling (HDM)
Economic Appraisal</td></tr>
<tr><td rowspan="2">D</td><td>Other Project Benefits
(Non motorised traffic)
(Producer Surplus Approach)</td><td>Optional</td><td>J</td><td>Sensitivity Analysis</td></tr>
<tr><td>Other Project Benefits
(Accidents)</td><td>Optional</td><td>K</td><td>Risk Analysis
(@RISK, Crystal Ball)</td></tr>
<tr><td>E</td><td>Vehicle Operating Costs</td><td>Mandatory</td><td colspan="2" rowspan="3"><u>Note</u>: the requirement for Risk Analysis is optional and will be dictated by the requirements of the Terms of Reference.</td></tr>
<tr><td>F</td><td>Physical Road Characteristics</td><td>Mandatory</td></tr>
<tr><td>G</td><td>Road Maintenance and Construction Costs</td><td>Mandatory</td></tr>
</table>

Traffic Range ADT (Average Daily Traffic: vehicles per day)	**Method of Analysis**	**Road Functional Type**		
		Inter-Urban (Primary, Secondary)	**Rural Feeder (Tertiary)**	**Urban, Peri-Urban (Single/Dual carriageway)**
ADT > 200	Vehicle Operating Cost Savings (HDM-4)	Y		
ADT 50 – 200	Vehicle Operating Cost Savings (RED)	Y	Y	
ADT < 50	Agricultural Producer Surplus	Y*	Y	
ADT > 1 000	Traffic Model (Land Use)			Y

* for a primary/secondary road, this would be supplementary to the primary (HDM-4 or RED) analysis

Appendix A

National Transport Master Plan (2004) –

Vehicle Operating Cost values

TABLE H-1
Standard Conversion Factor in USD 1,000

Breakdown	2000	2001	2002	Average
Imports	1,599,835	1,895,250	2,162,945	1,886,010
Exports	754,481	832,125	1,011,210	865,939
Taxes on Imports[1]	349,200	383,000	427,400	386,533
Subsidies of Exports	0	0	0	0
Standard Conversion Factor (SCF)	**0.87**	**0.88**	**0.88**	**0.88**

Notes 1: Exclude quasi government imports and VAT

TABLE H-2
Shadow Wage Rate

Breakdown	Unit	Skilled	Semi-skilled	Unskilled
Average Wage Rate [1]	USh per year	40,000,000	3,600,000	1,440,000
Cost of Employment [2]	USh per year	48,000,000	3,960,000	0
Average Production Loss[1]	USh per year	0	0	964,000
Income Tax Percentage[3]	%	30%	11%	0%
Unemployment Rate[4]	%	5%	10%	0%
Shadow Cost Factor	Factor	0.88	0.88	0.88
Economic Value of Labour Output	USh per year	28,089,600	2,791,325	848,320
Shadow Wage Factor		**0.70**	**0.78**	**0.59**

Sources:1. Reeves, J. - 2. Botes, F.J. - 3. Uganda Internal Revenue Service - 4. Hamilton, D.

TABLE H-3
Typical Motorised Vehicle Classes

HDM-4	Classification	Procedures Guide	MoWHC	JBG	Roughton	Gibb	Master Plan
0	Motorcycle		Honda 110cc	Honda 110cc	Yamaha 8100	Honda 110cc	Yamaha
1	Small Car						
2	Medium Car	Toyota Corona	Toyota Corolla 1.6	Toyota 1.6	Toyota Corolla 1.6	Toyota Corolla 1.6	Toyota Corolla 1.8
3	Large Car						
4	Light Delivery						
5	Light Goods (rural) Light Goods (urban)		Toyota Hilux	Toyota Hilux	Toyota Hilux 4X4 DC	Toyota Hilux	Nissan/Datsun 1.6 Toyota Hilux 4X4
6	Four Weal Drive		Mitshibushi Pajero	Mitshibushi Pajero	Toyota Landcruiser	Mitshibushi Pajero	Toyota Landcruiser
7	Light Truck	Toyota Dyna	Toyota Dyna	Toyota Dyna	Toyota Dyna	Toyota Dyna	Toyota Dyna
8	Medium Truck	Tata	Tata	Tata	Mitshibushi Canter	Tata	Tata
9	Heavy Truck	Mercedes	Mercedes	Mercedes	Isuzu	Mercedes	Mercedes
10	Articulated Truck	Mercedes	Mercedes	Mercedes	Nissan	Mercedes	Mercedes
11	Mini Bus (rural) Mini Bus (urban)	Toyota Hiace 2.4 diesel	Toyota Hiace 2.4 diesel	Toyota Hiace 1.8 petrol	Toyota Hiace	Toyota Hiace 1.8 petrol	Toyota Hiace 1.8 Toyota Hiace 2.4
13	Medium Bus						
14	Large Bus	Isuzu	Isuzu	Isuzu	Isuzu MV123	Isuzu	Isuzu

TABLE H-4
New and Reconditioned Vehicle Prices in USD

HDM-4 Classification		Vehicle Type	New	Reconditioned
0	Motorcycle	Yamaha	1,829	875
2	Medium Car	Toyota Corolla 1.8	27,025	4,250
5	Light Goods (urban)	Nissan/Datsun 1.6 petrol		4,500
	Light Goods (rural)	Toyota Hilux 4X4	26,487	
6	Four Weal Drive	Toyota Landcruiser Prado Std	43,841	11,500
7	Light Truck	Toyota Dyna/Isuzu Elf	22,440	7,000
8	Medium Truck	Tata	38,000	7,000
9	Heavy Truck	Mercedes	67,000	27,500
10	Articulated Truck	Mercedes	146,318	
11	Mini Bus (rural)	Toyota Hiace 1.8 petrol	28,739	5,250
	Mini Bus (urban)	Toyota Hiace 2.4 diesel	33,368	6,750
13	Medium Bus	Toyota	37,500	
14	Large Bus	Isuzu	56,000	

TABLE H-5
Financial and Economic Cost of New Vehicles in USD

Vehicle Type	Retail Price	VAT Percentage	VAT Amount	Import Duty Percentage	Import Duty Amount	Pre Tax Cost	Tyre Cost	Economic Cost Factor	Economic Cost
Motorcycle	1,829	17%	266	11.56%	162	1,401	47.93	0.81	1,353.07
Medium Car	27,025	17%	3,927	22.98%	4,284	18,645	159.76	0.69	18,485.24
Light Goods (rural)	26,487	17%	3,849	7.17%	1,499	20,912	379.42	0.79	20,532.58
Light Goods (urban)	17,000	17%	2,470	7.00%	860	13,670	219.67	0.80	13,450.33
Four Weal Drive	43,841	17%	6,370	21.26%	6,528	30,711	559.15	0.70	30,151.85
Light Truck	22,440	17%	3,270	7.00%	1,170	18,000	259.61	0.80	17,740.39
Medium Truck	38,000	17%	5,521	7.00%	1,935	30,544	894.64	0.80	29,649.36
Heavy Truck	67,000	17%	9,735	7.00%	3,746	53,519	2,416.33	0.78	51,102.32
Articulated Truck	146,318	17%	21,260	7.00%	8,181	116,877	7,492.61	0.76	109,384.39
Mini Bus (rural)	28,739	17%	4,176	14.00%	2,849	21,498	179.73	0.75	21,318.27
Mini Bus (urban)	33,368	17%	4,848	13.63%	3,394	24,910	179.73	0.75	24,730.27
Light Bus	37,500	17%	5,449	7.00%	1,800	30,251	259.61	0.81	29,991.39
Heavy Bus	56,000	17%	8,137	7.00%	3,375	44,488	1,537.66	0.79	42,950.34

TABLE H-6
Annual Vehicle Distance

HDM-4 Classification		RAFU	MASTER PLAN
0	Motorcycle	(*)	10,000
2	Medium Car	20,000	41,027
5	Light Goods (rural)	36,000	49,533
6	Four Weal Drive	(*)	50,136
7	Light Truck	40,000	65,810
8	Medium Truck	40,000	65,810
9	Heavy Truck	40,000	91,740
10	Articulated Truck	50,000	134,268
11	Mini Bus (rural)	(*)	70,908
14	Large Bus	60,000	135,396

Notes: (*) - No information

TABLE H-7
Financial and Economic Cost of Tyres

Vehicle Type	Tyre Type	Retail Price	VAT		Import Duty		Economic		Price - USD	
		(USh)	Percentage	Amount	Percentage	Amount	Cost	Cost Factor	Economic	Financial
Motorcycle		60,000	17%	8,718	7%	3,355	47,927	0.80	23.96	30.00
Medium Car	185/70 R 13	80,000	17%	11,624	7%	4,473	63,903	0.80	31.95	40.00
Light Goods (urban)	225/75 R15	190,000	17%	27,607	7%	10,624	151,769	0.80	75.88	95.00
Light Goods (rural)	195/80 R14	110,000	17%	15,983	7%	6,151	87,866	0.80	43.93	55.00
Four Weal Drive	265/70 R16	280,000	17%	40,684	7%	15,656	223,660	0.80	111.83	140.00
Light Truck	700 X 15	130,000	17%	18,889	7%	7,269	103,842	0.80	51.92	65.00
Medium Truck	900 X 20	320,000	17%	46,496	7%	17,893	255,611	0.80	127.81	160.00
Heavy Truck	1000 X 20	550,000	17%	79,915	7%	30,753	439,332	0.80	219.67	275.00
Articulated Truck	1200 X 20	670,000	17%	97,350	7%	37,463	535,187	0.80	267.59	335.00
Mini Bus (rural)	185/80 R14	90,000	17%	13,077	7%	5,032	71,891	0.80	35.95	45.00
Mini Bus (urban)	185/80 R14	90,000	17%	13,077	7%	5,032	71,891	0.80	35.95	45.00
Light Bus	700 X 15	130,000	17%	18,889	7%	7,269	103,842	0.80	51.92	65.00
Heavy Bus	1000 X 20	550,000	17%	79,915	7%	30,753	439,332	0.80	219.67	275.00

TABLE H-8
Tyre Wear Coefficient (TYRE_CTCE)

HDM-4 Classification		Default	Master Plan
0	Motorcycle	0.00050	0.00050
2	Medium Car	0.00187	0.00450
5	Light Goods (rural)	0.00187	0.00450
6	Four Weal Drive	0.00187	0.00450
7	Light Truck	0.00187	0.00250
8	Medium Truck	0.00201	0.01000
9	Heavy Truck	0.00275	0.01500
10	Articulated Truck	0.00311	0.02000
11	Mini Bus (rural)	0.00187	0.00450
14	Large Bus	0.00207	0.01000

TABLE H-9
Economic Price of Fuel

Breakdown	Unit	Economic Dealer Margin		Actual Dealer Margin	
		Petrol	Diesel	Petrol	Diesel
CIF Mombassa	USD per Ton	290	230	290	230
Litters / Ton	Litter	1,320	1,200	1,320	1,200
CIF	Per Litre	0.22	0.19	0.22	0.19
Piping Cost per Litre	Mombassa to Kisumu	0.04	0.04	0.04	0.04
Road Transport Cost per Litre	Kisumu to Kampala	0.03	0.03	0.03	0.03
CIF / Kampala	USD per Litre	0.29	0.26	0.29	0.26
Dealer and Retailer Costs and Margins	%	10%	10%	37%	27%
Actual Dealer and Retailer Costs and Margins	USD per Litre	0.03	0.03	0.11	0.07
Economic Cost of Fuel at Pump	USD per Litre	0.32	0.29	0.40	0.33
Economic Cost of Fuel at Pump	USh per Litre	640	578	797	667
Financial Retail Fuel Price	USD	1.37	0.75	1.37	0.75
Financial Retail Fuel Price	USh	2,740	1,490	2,740	1,490
Economic Cost Factor	Factor	0.23	0.39	0.29	0.45

TABLE H-10
Economic Price of Oil

Breakdown	Unit	Petrol	Diesel
Retail Price	USh per Litre	3,600	4,400
VAT	%	17%	17%
VAT	USh	523	639
Import Duty	%	15%	15%
Import Duty Amount	USh	401	491
Excise Duty	%	10%	10%
Excise Duty Amount	USh	243	297
CIF Price	USh per Litre	2,189	2,702
Assumed Dealer Cost and Margins	USh per Litre	243	270
Economic Cost	USh	2,432	2,972
Economic Cost	USh	0.68	0.68
Economic Cost	USD	1.22	1.49
Financial Cost	USD	1.8	2.2

TABLE H-11
Labour Hours Model Adjustment of Factor - PARTS_KP

HDM-4 Classification		Default	Master Plan
0	Motorcycle	0.308	0.308
2	Medium Car	0.260	0.260
5	Light Goods (rural)	0.308	0.308
6	Four Weal Drive	0.371	0.308
7	Light Truck	0.371	0.308
8	Medium Truck	0.371	0.308
9	Heavy Truck	0.371	0.308
10	Articulated Truck	0.371	0.308
11	Mini Bus (rural)	0.308	0.308
14	Large Bus	0.483	0.483

TABLE H-12
Labour Hours Model Adjustment of Factor - LAB_A0

HDM-4 Classification		Default	Master Plan
0	Motorcycle	77.14	77.14
2	Medium Car	77.14	77.14
5	Light Goods (rural)	77.14	77.14
6	Four Weal Drive	77.14	77.14
7	Light Truck	242.03	77.14
8	Medium Truck	301.46	77.14
9	Heavy Truck	301.46	77.14
10	Articulated Truck	301.46	77.14
11	Mini Bus (rural)	77.14	77.14
14	Large Bus	293.44	77.14

TABLE H-13
Breakdown of Driver and Crew Cost per Hour

Vehicle Type	Unit	Motorcycle	Medium Car	Light Goods (Rural)	Light Goods (Urban)	Four Weal Drive	Light Truck	Medium Truck	Heavy Truck	Articulated Truck	Mini Bus (Rural)	Mini Bus (Urban)	Medium Bus
DRIVER													
Average Number of Paid Drivers		0.64	0.45	0.85	0.85	0.79	0.82	0.86	0.89	1.00	1.00	1.00	1.00
Average Hourly Wage for Driver		619	1199	1199	1199	1199	1199	1199	1199	1199	1199	1199	1199
Average Hourly Driver Cost		396	540	1019	1019	947	983	1031	1067	1199	1199	1199	1199
SWR for Driver		0.59	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78
Economic Driver Cost	USh	234	421	795	795	739	767	804	832	935	935	935	935
Economic Driver Cost	USD	0.12	0.21	0.40	0.40	0.37	0.38	0.40	0.42	0.47	0.47	0.47	0.47
Financial Driver Cost	USD	0.20	0.27	0.51	0.51	0.47	0.49	0.52	0.53	0.60	0.60	0.60	0.60
VEHICLE SUPPORT													
Average Number of Paid Support Staff		0.00	0.00	0.17	0.17	0.64	2.05	2.92	1.42	1.70	1.20	1.20	2.00
Average Hourly Wage for Support		619	619	619	619	619	619	619	619	619	619	619	619
Average Hourly Support Staff Cost		0	0	105	105	396	1268	1806	878	1052	742	742	1237
SWR for Crew		0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59
Economic Support Staff Cost	USh	0	0	62	62	234	748	1066	518	620	438	438	730
Economic Support Staff Cost	USD	0.00	0.00	0.03	0.03	0.12	0.37	0.53	0.26	0.31	0.22	0.22	0.36
Financial Support Staff Cost	USD	0.00	0.00	0.05	0.05	0.20	0.63	0.90	0.44	0.53	0.37	0.37	0.62
TOTAL CREW COST													
Total Economic Cost	USh	234	421	857	857	972	1515	1870	1351	1556	1373	1373	1665
Total Economic Cost	USD	0.12	0.21	0.43	0.43	0.49	0.76	0.93	0.68	0.78	0.69	0.69	0.83
Financial Cost	USD	0.20	0.27	0.56	0.56	0.67	1.13	1.42	0.97	1.13	0.97	0.97	1.22

	Days	Hours	Vehicle Service Wage	Financial	Economic	Skilled
Year	365	2,920	Franchise	9,309	7,261	12,066
Weekend	52	416	Non-Franchise	1,307	1,019	
Public Holiday	10	80				
Leave	12	96				
		2,328				

TABLE H-14
Vehicle Operating Overhead Costs

Vehicle Type	Average Vehicle Value	Annual Fees	Insurance Cost	Total Overhead Costs
Motorcycle	1,353	30	41	71
Medium Car	18,485	142	277	419
Light Goods (urban)	20,533	162	308	470
Light Goods (rural)	13,450	162	202	364
Four Weal Drive	30,152	312	452	764
Light Truck	17,740	162	266	428
Medium Truck	29,649	212	445	657
Heavy Truck	51,102	262	767	1,029
Articulated Truck	109,384	312	1,641	1,953
Mini Bus (rural)	21,318	242	640	882
Mini Bus (urban)	24,730	242	742	984
Light Bus	29,991	398	900	1,297
Heavy Bus	42,950	553	1,289	1,842

TABLE H-15
Constants for Calculating VOC

Fuel Consumption	Kc	Kd	Ke	Kf
Cars	100.6	1467.6	-1.701	0.01524
Light Goods Vehicle	62.6	2526.2	-1.1937	0.01465
Light Truck	140.6	2588.3	-2.128	0.02362
Medium Truck	475.5	5632.4	-10.753	0.10156
Heavy Truck	258	14597.7	-9.432	0.12023
Minibuses	-60.9	4259.8	1.846	-0.00339
Bus	295.7	7045.7	-5.338	0.06359

Oil Consumption	Ro	B
Cars	440	0.0028
Light Goods Vehicle	450	0.0028
Light Truck	1010	0.0021
Medium Truck	2750	0.0021
Heavy Truck	3060	0.0021
Minibuses	480	0.0028
Bus	2450	0.0021

Tyre Consumption	Ki	Kj	Kk	Kl	Km
Cars	-0.07634	0.871989	0.5	0.11833162	0.02
Light Goods Vehicle	-0.07634	0.871989	0.5	0.11833162	0.02
Light Truck	-0.03011	1.897469	0.2	0.00761221	0.03
Medium Truck	-0.03575	2.089001	0.2	0.01038345	0.03
Heavy Truck	-0.03575	2.089001	0.2	0.01038345	0.03
Minibuses	-0.07634	0.871989	0.5	0.11833162	0.02
Bus	-0.03011	1.897469	0.2	0.00761221	0.03

Maintenance	Kq	Kr	Ks
Cars	3.342057	1.015391	-0.92333
Light Goods Vehicle	3.480565	1.019512	-0.91042
Light Truck	1.71553	1.01754	-0.79752
Medium Truck	1.405382	1.016991	-0.84421
Heavy Truck	0.989257	1.020283	-0.85948
Minibuses	3.668089	1.01697	-0.89395
Bus	1.366491	1.02111	-0.90174

Depreciation	Kn	Ko	Kp
Cars	-0.46542	0.523317	0.306642
Light Goods Vehicle	-0.67941	0.685873	0.238187
Light Truck	-1.02666	1.017971	0.287964
Medium Truck	-1.13363	1.127412	0.276091
Heavy Truck	-1.75486	1.986278	0.409804
Minibuses	-1.06463	1.0728	0.242181
Bus	-2.53144	2.323725	0.24985

TABLE H-26
Value of Passenger Travel Time in USD

	HDM-4 Classification	Work Hour	Non - Work Hour
0	Motorcycle	0.47	0.12
2	Medium Car	7.80	1.95
5	Light Goods (r)	0.18	0.18
6	Four Weel Drive	7.80	7.80
7	Light Truck	0.18	0.18
8	Medium Truck	0.18	0.18
9	Heavy Truck	0.18	0.18
10	Articulated Truck	0.18	0.18
11	Minibus(r)	0.18	0.18
14	Large Bus	0.47	0.12

TABLE H-27
Vehicle Passenger Usage

	HDM-4 Classification	% Private Trips	Passengers per Vehicle	Percentage Private Usage
0	Motorcycle	75.00	1	15.00
2	Medium Car	81.00	3	38.00
5	Light Goods (r)	82.00	3	38.00
6	Four Weel Drive	81.00	3	38.00
7	Light Truck	0.00	0	58.00
8	Medium Truck	0.00	0	58.00
9	Heavy Truck	0.00	0	58.00
10	Articulated Truck	0.00	0	58.00
11	Minibus(r)	0.00	13	46.00
14	Large Bus	0.00	53	43.00

Appendix B

Road Roughness values

Overseas Road Note (ORN) 5 – 1988
Table 10.2

World Bank Technical Paper Number 46

Guidelines for Conducting and Calibrating Road Roughness Measurements

(extract from Chapter 5)

TABLE 10.2

ROAD ROUGHNESS VALUES FOR DIFFERENT ROAD SERVICEABILITY LEVELS

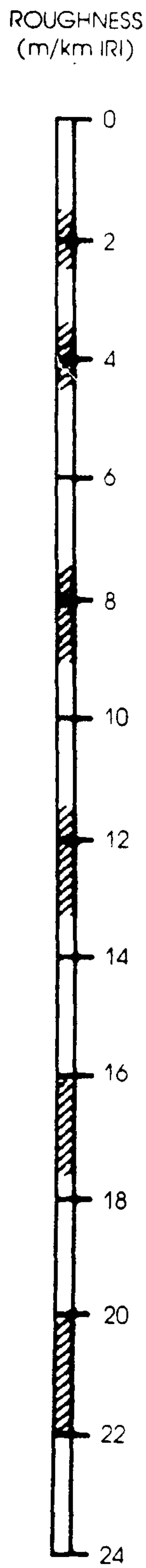
Paved roads

Serviceability description	m/km IRI	mm/km BI
Ride comfortable over 120 km/h. Undulation barely perceptible at 80 km/h in range 1.3 to 1.8. No depressions, pot-holes or corrugations are noticeable: depressions <2 mm/3 m. Typical high quality asphalt 1.4 to 2.3. High quality surface treatment 2.0 to 3.0.	1.5–2.5	1000–2000
Ride comfortable up to 100–200 km/h. At 80 km/h, moderately perceptible movements or large undulations may be felt. Defective surface: occasional depressions, patches or pot-holes (eg 5–15 mm/3 m or 10–20 mm/5 m with frequency 1–2 per 50 m) or many shallow pot-holes (eg on surface treatment showing extensive ravelling). Surface without defects: moderate corrugations or large undulations.	4.0–5.5	3000–4000
Ride comfortable up to 70–90 km/h, strongly perceptible movements and swaying. Usually associated with defects: frequent moderate and uneven depressions or patches (eg 15–20 mm/3 m or 20–40 mm/5 mm with frequency 5–3 per 50 m) or occasionally pot-holes (eg 1–3 per 50 m). Surface without defects: strong undulations or corrugations.	7.0–8.0	5500–6500
Ride comfortable up to 50–60 km/h, frequent sharp movements or swaying. Associated with severe defects: frequent deep and uneven depressions and patches (eg 20–40 mm/3 m or 40–80 mm/5 m with frequency 5–3 per 50 m) or frequent pot-holes (eg 4–6 per 50 m).	9.0–10.0	7000–8000
Necessary to reduce velocity below 50 km/h. Many deep depressions pot-holes and severe disintegration (eg 40–80 mm deep with frequency 8–16 per 50 m).	11.0–12.0	9000–10000

Unpaved roads

Serviceability description	m/km IRI	mm/km BI
Recently bladed surface of fine gravel, or soil surface with excellent longitudinal and transverse profile (usually found only in short lengths).	1.5–2.5	1000–2000
Ride comfortable up to 80–100 km/h. Aware of gentle undulations or swaying. Negligible depressions (eg <5 mm/3 m) and no pot-holes.	3.5–4.5	2500–3500
Ride comfortable up to 70–80 km/h but aware of sharp movements and some wheel bounce. Frequent shallow-moderate depressions or shallow pot-holes (eg 6–30 mm/3 m with frequency 5–10 per 50 m). Moderate corrugations (eg 6–20 mm/0.7–1.5 m).	7.5–9.0	6000–7000
Ride comfortable at 50 km/h (or 40–70 km/h on specific sections). Frequent moderate transverse depressions (eg 20–40 mm/3–5 m at frequency 10–20 per 50 m) or occasional deep depressions or pot-holes (eg 40–80 mm/3 m with frequency less than 5 per 50 m). Strong corrugations (eg <20 mm/0.7–1.5 m).	11.5–13.5	9500–11500
Ride comfortable at 30–40 km/h. Frequent deep transverse depressions and/or pot-holes (eg 40–80 mm/1–5 m with frequency less than 5 per 50 m) with other shallow depressions. Not possible to avoid all the depressions except the worst.	16.0–17.5	14000–15500
Ride comfortable at 20–30 km/h. Speeds higher than 40–50 km/h would cause extreme discomfort, and possible damage to the car. On a good general profile: frequent deep depressions and/or pot-holes (eg 40–80 mm/1–5 m at frequency 10–15 per 50 m) and occasional very deep depressions (eg <80 mm/0.6–2 m). On a poor general profile: frequent moderate defects and depressions (eg poor earth surface).	20.0–22.0	18000–20000

UNPAVED ROADS (GRAVEL/EARTH)



Recently bladed surface of fine gravel, or soil surface with excellent longitudinal and transverse profile (usually found only in short lengths).

Ride comfortable up to 80-100 km/h, aware of gentle undulations or swaying. Negligible depressions (e.g. < 5mm/3m) and no potholes.

Ride comfortable up to 70-80 km/h but aware of sharp movements and some wheel bounce. Frequent shallow-moderate depressions or shallow potholes(e.g. 6-30mm/3m with frequency 5-10 per 50m). Moderate corrugations (e.g. 6-20mm/0.7-1.5m).

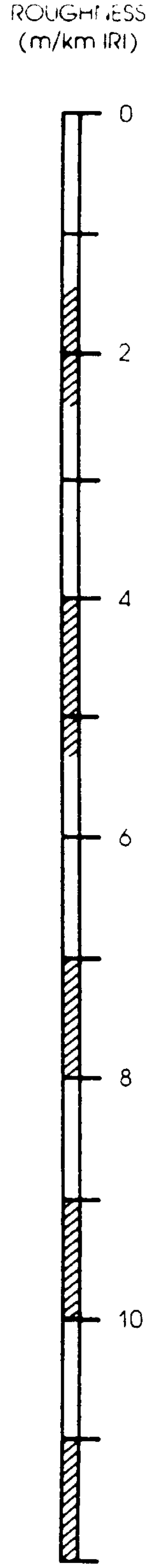
Ride comfortable at 50km/h (or 40-70 km/h on specific sections). Frequent moderate transverse depressions (e.g. 20-40mm/3-5m at frequency 10-20 per 50m) or occasional deep depressions or potholes (e.g. 40-80mm/3m with frequency less than 5 per 50m). Strong corrugations (e.g. > 20mm/0.7-1.5m).

Ride comfortable at 30-40 km/h. Frequent deep transverse depressions and/or potholes (e.g. 40-80mm/1-5m at freq. 5-10 per 50m); or occasional very deep depressions (e.g. 80mm/1-5m with frequency less than 5 per 50m) with other shallow depressions. Not possible to avoid all the depressions except the worst.

Ride comfortable at 20-30 km/h. Speeds higher than 40-50 km/h would cause extreme discomfort, and possible damage to the car. On a good general profile: frequent deep depressions and/or potholes (e.g. 40-80mm/1-5m at frequency 10-15 per 50m) and occasional very deep depressions (e.g. > 80mm/0.6-2m). On a poor general profile: frequent moderate defects and depressions (e.g. poor earth surface).

Fig. 12. Road roughness estimation scale for paved roads with asphaltic concrete or surface treatment (chipseal) surfacings. .

PAVED ROADS (ASPHALTIC CONCRETE/SURFACE TREATMENT)



Ride comfortable over 120 km/h. Undulation barely perceptible at 80 km/h in range 1.3 to 1.8. No depressions, potholes or corrugations are noticeable; depressions < 2mm/3m. Typical high quality asphalt 1.4 to 2.3, high quality surface treatment 2.0 to 3.0

Ride comfortable up to 100-120 km/h. At 80 km/h, moderately perceptible movements or large undulations may be felt. Defective surface: occasional depressions, patches or potholes (e.g. 5-15mm/3m or 10-20mm/5m with frequency 2-1 per 50m), or many shallow potholes (e.g. on surface treatment showing extensive ravelling). Surface without defects: moderate corrugations or large undulations.

Ride comfortable up to 70-90km/h, strongly perceptible movements and swaying. Usually associated with defects: frequent moderate and uneven depressions or patches (e.g. 15-20mm/3m or 20-40mm/5m with frequency 5-3 per 50m), or occasional potholes (e.g. 3-1 per 50m). Surface without defects: strong undulations or corrugations.

Ride comfortable up to 50-60 km/h, frequent sharp movements or swaying. Associated with severe defects: frequent deep and uneven depressions and patches (e.g. 20-40mm/3m or 40-80mm/5m with frequency 5-3 per 50m), or frequent potholes (e.g. 4-6 per 50m).

Necessary to reduce velocity below 50km/h. Many deep depressions, potholes and severe disintegration (e.g. 40-80 mm deep with frequency 8-16 per 50m).

Fig. 13. Road roughness estimation scale for unpaved roads with gravel or earth surfaces.